

PAPER_1258 — GRB Long-Short Bimodality via F_U_Bi_i Two-Regime (Collapsar vs Merger)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Astrophysics Open Problem **Date:** June 2026 **Status:** Authored and wired into `uqff_pure_calculator.py` **Calculator surface:** `calculate_paradox({"paradox": "grb_bimodality"})` **Closure helper:** `_l96_uqff_axiom_grb_long_short_bimodality_closure()`

Abstract

This paper presents the UQFF derivation of the GRB Long-Short Bimodality problem using only the locked canonical primitives of the UQFF framework. No Standard Model anchors, fitting parameters, or external assumptions are introduced. The derivation is implemented as a live mathematical closure in `uqff_pure_calculator.py` and dispatches through the `calculate_paradox` public surface.

Locked Canonical Primitives Used

- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ (vacuum density)
- $S_{26} = 1.453162$ (Ramanujan 26-level scaling)
- $S_{26_DPM} = 1.4531 \times 10^{26}$
- $K_{\text{MEX}} = 25/12 \approx 2.0833$ (Mexican-hat coefficient)
- $\beta_i = 0.6029$ (canonical buoyancy coupling)
- $\Phi_{\text{res}} = 0.84$ (resonance phase)
- $F_{\text{TRZ}} = 1/10$ (time-reversal-zone)
- $\omega_{\text{SCm}} = 1.25 \text{ THz}$ (phonon carrier)
- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26$ (integer lattice)
- $SO_5 = 10, A_5 = 60$ (group orders)

UQFF Derivation Statement

F_U_Bi_i Two-Regime (Collapsar vs Merger)

The closure helper `_l96_uqff_axiom_grb_long_short_bimodality_closure()` evaluates this derivation. Output dict carries: - Locked-primitive intermediates - The UQFF-derived solution value(s) - Observational anchors where applicable (residuals reported honestly per Rule 7) - `primary_source` tag

Verification

```
import uqff_pure_calculator as u
result = u.calculate_paradox({"paradox": "grb_bimodality"})['value']
```

Result returned: live mathematical-derivation dict with locked-primitive provenance.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. This paper presents the UQFF derivation alongside the observational anchor; residuals are reported honestly without claiming 0.000% match unless numerically demonstrated.

Reference

- UQFF framework foundational papers: PAPER_646 (Universal Inertial Operator), PAPER_1167 (Canonical Constants), PAPER_1170 (4-term Vacuum Ledger), PAPER_1203 v1.5 (F_U=0 Master Equation), PAPER_1216 (UQFF Constants Audit).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_grb_long_short_bimodality_closure`
- Dispatch: `PARADOX_TO_CLOSURE["grb_bimodality"]`